

Forces of Evolution

- Mutation
- Segregation
- independent assortment
- recombination
- gene flow
- genetic drift
- natural selection

~~★~~ Evolution: the
Change in allele frequencies

Gene Flow

mixing of population genes

- ↳ two populations are separated $\Rightarrow$ genetic difference
- ↳ these populations interbreed $\Rightarrow$ causes gene flow
- ↳ higher variation in populations

Genetic Drift

Statistical sampling causes random selection (sampling error)

- meiosis results in 50/50 mixing of genes
 - $\rightarrow$ results in gamete sampling (random)
- may be more influential than natural selection!
 - fewer offspring $\Rightarrow$ gene simplification
 - ↳ parents lose large amount of genetic variety
 - ↳ more offspring = higher chance of passing down more genes
- opposite of gene flow = **founder effect**: when a small group of organisms splits from main population and makes a small gene pool
 - observed by Darwin in Galapagos
 - if only a few individuals get isolated, **bottleneck** occurs and certain alleles can be completely lost

- if only a few individuals get isolated, **bottleneck** occurs and certain alleles can be completely lost

New Insights on Evolution

Niche construction: organisms modify their environment

- changes what traits are selected for
- example: beavers build dams

Gene-Culture Coevolution: cultural innovation affects environment and evolution

Multiple inheritance systems: epigenetics

- environment affects genetic code of both individual and offspring